Where Data Lives and the Data Pipeline

BY THE END:

You can name the stages a dataset passes through, and where it is stored at each one.

Overview

The sales figure on a manager's dashboard this morning began three weeks ago as a digital wallet payment for a sack of rice. Between those two moments it was moved, stored, cleaned, reshaped, and summarized, usually by systems nobody in that meeting has ever seen. Data does not exist in one single place. It moves through systems and stages before it becomes useful.

Where data lives

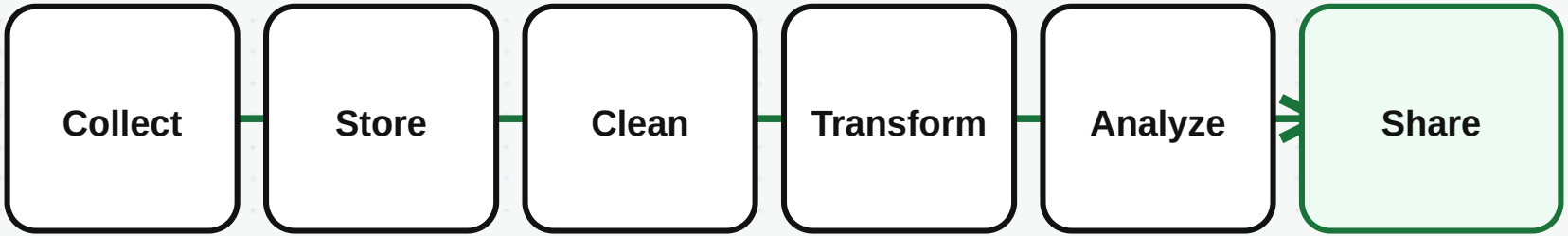
Organizations store data in different systems depending on purpose and scale.

SYSTEM	PURPOSE
Database	Stores and retrieves structured business data for daily operations
Data warehouse	Stores large amounts of historical data for analysis
Data lake	Stores raw data in many formats, often at large scale
Spreadsheets / files	Useful for small datasets and quick work

The difference that matters is what each one is *for*. A database is built to record what is happening right now, one transaction at a time. A warehouse is built to answer questions about the last three years. Ask a database a three-year question and you will slow down the shop it is running.

What is a data pipeline?

A **data pipeline** is the path data follows from its source to its final use.



Each stage has a purpose:

Collect - gather data from apps, forms, sensors, or systems

Store - save it in a database, warehouse, or file

Clean - fix missing values, duplicates, and errors

Transform - reshape data into a useful format

Analyze - explore patterns and answer questions

Share - present results in reports or dashboards

Example

Follow the payment from the top of the lesson.

1. The digital wallet system **collects** the payment - amount, time, shop
2. It is **stored** in the payment provider's database, alongside millions of others
3. A pipeline **cleans** the records, dropping the reversed and duplicated ones

4. The data is **transformed** into daily totals per shop
5. An analyst **analyzes** which shops are growing and which are flat
6. The manager sees the result in a **dashboard**, three weeks later

Six stages, and the number on the dashboard is the only one anybody in that meeting will ever see.

